Chapter 12, Problem 85CP

Problem

Design a three-phase heater with suitable symmetric loads using wye-connected pure resistance. Assume that the heater is supplied by a 240-V line voltage and is to give 27 kW of heat.

Step-by-step solution

Step 1 of 1

Let $Z_y = R$

$$V_P = \frac{V_L}{\sqrt{3}} = \frac{240}{\sqrt{3}}$$

$$= 138.56\ V$$

$$P = V_P I_P = \frac{27}{2} = 9\ KW = \frac{V_P^2}{R}$$

$$\Rightarrow R = \frac{V_P^2}{P} = \frac{(138.56)^2}{9000}$$

$$= 2.133\ \Omega$$

Thus,

$$Z_y = 2.133\ \Omega\ .$$